Foliations, Serre–Swan and Tepui Fibrations

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In [3], we introduce tepui fibrations, a class of diffeological fibrations that models the smooth structure of holonomy groupoids of singular foliations. This yields a Serre-Swan correspondence for locally finitely generated $C^\infty(M)$-modules and provides a natural framework for singular Lie theory.

1. INTRO: a crash course on Diffeology

DIFFEOLOGICAL SPACES: For a diffeological space X smoothness is defined by defining

which maps $U \subset \mathbb{R}^n \to X$ are smooth.

Such maps are called plots. The collection of all plots is usually denoted $\mathcal{D}$ and it satisfies topological/sheaf-like axioms. We say $\mathcal{D}$ is a diffeology.

A *diffeological space* is a pair $(X, \mathcal{D})$ where X is a set and $\mathcal{D}$ is a diffeology on X.

MAPS: A map between diffeological spaces $f\colon (X,\mathcal{D}) \to (Y, \mathcal{D}')$ is called *smooth if* $f \circ \mathcal{D} \subseteq \mathcal{D}'$.

QUOTIENTS: *Any quotient* $Y = (X/\sim)$ *of a diffeological space* $(X,\mathcal{D})$ *has a diffeology, namely the smallest one making the quotient map, a smooth map. This means a map* $U \subset \mathbb{R}^n \to Y$ *is a plot if and only if one can locally lift it to a plot of* X.

PLOTWISE-SUBMERSION: • *Maps that behave like quotients are called subductions. This is* $X \to Y$ *is a subduction if for any plot* $U \subset \mathbb{R}^n \to Y$ *there exists a lifting plot* $U \subset \mathbb{R}^n \to X$.

- **Plotwise-submersions** *are subductions onto their images and such that lifting plots can be arranged to pass through any element in* X *of our choosing. Plotwise-submersions work similar to submersions.*

2. MAIN OBJECT: Tepui fibrations

TEPUI FIBRATIONS: A tepui fibration is $s\colon X \to M$, a diffeological space X with a surjective plotwise submersion s to a manifold M such that:

$$(\forall x \in X)\ \exists\ \left(k_x \in \mathbb{N}\,,\ U_x \subset_\circ \mathbb{R}^{k_x}\,,\ q_x\colon U_x \to X\right)$$

satisfying:

1. Fibers of s are Hausdorff and second countable, .
2. $x \in \operatorname{Im}(q_x)$ and $q_x\colon U_x \to X$ are plotwise submersions (they work like quotients). This naturally implies that, the map $s \circ q_x\colon U_x \to M$ is a (manifold) **submersion**, and has smooth fibers.
3. Restricting the maps to the fibers $q_x\colon (U_x)_{p(x)} \to X$ gives **injectivity** (Debord's property).

Proposition 2.1. *If* $s\colon X \to M$ *is a tepui fibration then:*

- *The* s *fibers are smooth manifolds, the dimension of the manifold depends on the points in* M.
- *if the maps* $q_x\colon U_x \to X$ *have connected fibers then they are quotients by singular distribution in* $U_x \subset_\circ \mathbb{R}^{k_x}$.

Example 2.2. *Given a vector bundle* $E \to M$ *and a singular distribution* $V \to M$ *the quotient* $p\colon (E/V) \to M$ *is a tepui fibration.*

About the name, tepuis are a type of inselberg, isolated mountains on plains formed by erosion of an older highland. In this picture U_x =original highland, X =tepui-mountain and M =plains. Tepuis are also the setting of The Lost World by Arthur Conan Doyle and inspired the landscape in Pixar's Up.

3. MOTIVATION: holonomy groupoid of a singular foliation

SINGULAR FOLIATIONS: A *singular foliation* on a smooth manifold, in the sense of [1], is a $C^\infty(M)$-module $\mathcal{F}$ of tangent vector fields which is:

- involutive, i.e., $[\mathcal{F},\mathcal{F}] \subset \mathcal{F}$, and
- locally finitely generated.

Singular foliations are naturally induced by **Lie algebra actions** on manifolds, **Poisson structures**, **Lie algebroid**s, and much more.

For the construction of the holonomy groupoid (as in [1]), let us fix a singular foliated manifold $(M,\mathcal{F})$. By definition of locally finitely generated:

$$(\forall\ p \in M)\ \exists\ \left(\ k_p \in \mathbb{N}\ ,\ U_p \subset M\ ,\ X_1, \dots X_{k_p} \in \mathfrak{X}(U_p)\ \right) \quad \text{s.t.}$$

$$\mathcal{F}|_{U_p} = \operatorname{Span}(X_1, \dots, X_{k_p}) \quad \text{minimally.}$$

With the above data there are Lie groupoid-like objects for every $p \in M$:

$$\left(\mathbb{R}^{k_p} \times U_p, (0,p)\right) \rightrightarrows M$$

with source $s = \operatorname{Pr}_{U_p}$ and target t given by the flows of $X_i's$.

HOLONOMY GROUPOID: The holonomy groupoid (Androulidakis-Skandalis in [1]) is (near the identity bisection) a quotient

$$\left(\left(\sqcup_{p\in M}\left(\mathbb{R}^{k_p} \times U_p, (0,p)\right)\right)/\sim\right) =: \mathcal{U} \subset \mathcal{H}(\mathcal{F})$$

$\forall p \in M$ the quotient maps:

$$\left(\mathbb{R}^{k_p} \times U_p, (0,p)\right) \xrightarrow{Q_p} \mathcal{U}$$

satisfy that:

1. They are a **plotwise submersions** (see Prop 2.10 [1]). Implying that for any $p, q \in M$ with $\operatorname{Im}(Q_p) \cap \operatorname{Im}(Q_q) \neq \emptyset$ there will be **change of coordinates**.
2. Near $(0,p) \in \mathbb{R}^{k_p} \times U_p$ the map $Q_p|_{\mathbb{R}^{k_p}\times\{p\}}$ **is locally injective** on each source fibre (See Prop 1.1 [2] by Debord).

This means that the source map $s\colon \mathcal{H}(\mathcal{F}) \to M$ **is a tepui fibration.**

4. MAIN THEOREM: Serre-Swan and VB-tepui

VB-TEPUI: A bundle of vector spaces $p\colon W \to M$ over a manifold M, is called *VB-tepui* if p is a tepui fibration.

Proposition 4.1. *Any VB-tepui* $W \to M$ *is locally a quotient of vector bundles* E/V *as in ex. 2.2.*

Proof. idea or construction: For $p \in M$ take $e_1, \dots, e_k$ a basis for W_p, using that $W \to M$ is a local subduction we can find sections $\sigma_i\colon (M,p) \mapsto (W, e_i)$. Let U the intersection of the domain of all σ_i. The map:

$$Q\colon \mathbb{R}^k \times U \to W; (\alpha_1, \dots, \alpha_k, u) \mapsto \sum_i \alpha_i \sigma_i(u)$$

is a local subduction (possibly reducing U) and a vector bundle morphism, taking $E = \mathbb{R}^k \times U$ and $V = \ker(Q)$ we get as desired. □

Theorem 4.2. *(Serre-Swan-type) There is an equivalence of categories between VB-tepuis and locally finitely generated* $C^\infty(M)$*-modules (up to invisible elements).*

$$\boxed{\text{VB-tepuis} \ \leftrightarrow\ C^\infty(M)\text{-Mod}_{\text{lfg}}}$$

Proof. idea: One can see that the sections of a VB-tepui are indeed locally finitely generated modules.
Then given a locally finitely generated module $\mathcal{F}$, and a point $p \in M$ choose local generators $\sigma_1, \dots, \sigma_k$, a map similar to Q in the last proposition gives a VB-tepui whose sections are isomorphic to the elements of $\mathcal{F}$ near p.
Finally we can prove that this construction is functorial, i.e. $C^\infty(M)$-morphisms gives diffeological vector bundle morphisms and vice versa. □

5. TEPUI ALGEBROIDS

Definition 5.1. *A (Lie) tepui algebroid is a triple* $(A \to M, \rho, [\cdot,\cdot])$ *where:*

- $A \to M$ *is VB-tepui;*
- $\rho\colon A \to TM$ *is a VB-tepui morphism over* 1_M*. We call* ρ *the anchor;*
- $[\cdot,\cdot]\colon \Gamma(A) \times \Gamma(A) \to \Gamma(A)$ *is a Lie bracket (i.e., bilinear, skew-symmetric and satisfies the Jacobi identity);*

such that the following Leibniz identity holds:

for all $a, b \in \Gamma(A)$, *and for all* $f \in C^\infty(M)$, *we have* $[a, fb] = f[a,b] + \rho(a)(f)b$

In view of the singular Serre-Swan Theorem 4.2, we can give a completely algebraic characterization of tepui algebroids: a tepui algebroid is a locally finitely generated Lie-Rinehart algebra over $C^\infty(M)$ (up to invisible elements).

Example 5.2 (Tepui algebroids from singular foliations)**.** *Let* $\mathcal{F}$ *be a singular foliation. Then there is a VB-tepui* A *such that* $\Gamma(A) \cong \mathcal{F}$*. To obtain it, we apply Theorem 4.2 to* $\mathcal{F}$.

Proposition 5.3. *If* $(A, \rho, [\cdot,\cdot])$ *is a tepui algebroid then* $\rho(\Gamma_c(A)) \subset \mathfrak{X}_c(M)$ *is a singular foliation.*

Other structures inducing a tepui algebroids are:

- Singular subalgebroids as in [6].
- Almost Lie algebroids as in [4].
- Lie ∞-algebroids as in [5].

6. WORK IN PROGRESS: Tepui groupoids

TEPUI GROUPOIDS: A *tepui groupoid* is a diffeological groupoid over a manifold $\mathcal{H} \rightrightarrows M$ such that the source map $s\colon \mathcal{G} \to M$ (and therefore also the target map) is a tepui fibration (+ one extra property).

Example 6.1. *The holonomy groupoid of a singular foliation is a tepui Groupoid.*

Proposition 6.2. *If* $\mathcal{H} \rightrightarrows M$ *is a tepui groupoid then the set of left invariant vector fields on* $\mathcal{G}$ *is a locally finitely generated* $C^\infty(M)$ *module and a Lie-Rhineart algebra. The corresponding VB-tepui is automatically a tepui Algebroid.*

Conjecture 6.3. *There is a 1-1 correspondence between singular foliations and effective tepui groupoids.*

References

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